

CMSC 207: Chapter 10 Homework

February 16, 2026

1. Prove that having n vertices (where $n \in \mathbb{Z}^+$) is an invariant for a **graph isomorphism**.
-

Let G and H be graphs, and let G be isomorphic to H .

Let V_G be the vertex set of G and V_H be the vertex set of H .

Let n be the number of vertices in G , so $|V_G| = n$.

By the definition of a graph isomorphism, there exists a bijection $f : V_G \rightarrow V_H$.

Since f is a bijection, it is both one-to-one and onto.

- Since f is one-to-one, no two vertices in G can map to the same vertex in H . Thus, the number of vertices in H must be at least the number of vertices in G : $|V_H| \geq |V_G|$.
- Since f is onto, every vertex in H must correspond to some vertex in G . Thus, the number of vertices in H cannot exceed the number of vertices in G : $|V_H| \leq |V_G|$.
- Combining the two inequalities, we have $|V_H| = |V_G|$. If G and H are isomorphic, then the number of vertices in G must be equal to the number of vertices in H .

Since $|V_G| = n$, then $|V_H| = n$. Thus, having n vertices is an invariant for a graph isomorphism. ■

2. Prove that the sum of degrees of the vertices of any finite graph is even.

Let $G = (V, E)$ be a finite graph.

Each edge $e \in E$ has either one or two endpoints:

- If e has one endpoint, then it contributes 2 to the sum of degrees, since it is incident to the same vertex twice.
- If e has two endpoints u and v , then it contributes 1 to the degree of u and 1 to the degree of v , for a total contribution of 2 to the sum of degrees.

Since each edge in E contributes 2 to the sum of degrees, the total sum of degrees of all vertices in G is $2|E|$. Since $|E| \in \mathbb{Z}$, then $2|E|$ is even.

Thus, the sum of degrees of the vertices of any finite graph is even. ■

3. Show that every simple finite graph has two vertices of the same degree.

Note: A simple graph is an undirected graph with no loops and no parallel edges.

Let $G = (V, E)$ be a simple finite graph with n vertices.

Since G is simple, a vertex in G cannot be adjacent to itself, and it cannot be adjacent to another vertex through more than one edge.

- That is, a vertex in G can only be adjacent to at most $n - 1$ other vertices.
- Thus, the degree of any vertex v in G must satisfy $0 \leq \deg(v) \leq n - 1$.

A vertex with degree 0 cannot coexist with a vertex with degree $n - 1$, since the vertex with degree $n - 1$ would have to be adjacent to every other vertex, including the vertex with degree 0, which is a contradiction.

- Thus, the possible degrees of the vertices in G are either $0, 1, 2, \dots, n - 2$ or $1, 2, \dots, n - 1$. In both cases, there are only $n - 1$ possible degrees for n vertices.

By the pigeonhole principle, since there are more vertices than possible degrees, at least two vertices in G must take on the same degree. ■